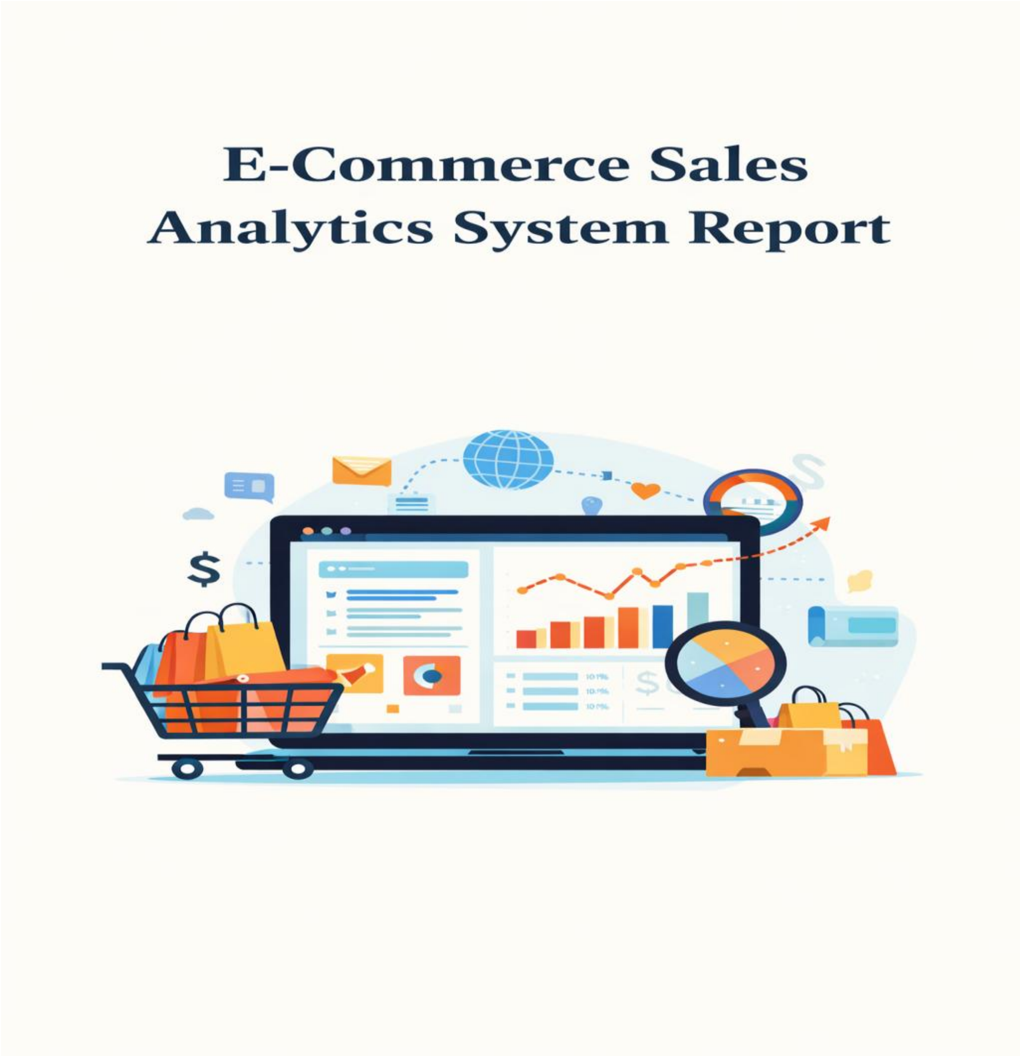
E-Commerce Sales Analytics System Report

SALES ANALYSIS



E-Commerce Sales Analytics Using Power Query &SQL

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Overview

This project focuses on analyzing e-commerce transactional sales data using SQL to extract meaningful business insights. The dataset includes customer information, product details, order records, shipping data, and financial metrics such as quantity, discounts, revenue, and profit.

The raw data was cleaned, validated, and transformed into a normalized relational database model to support efficient querying and accurate reporting. The structured model enables businesses to evaluate product performance, customer purchasing behavior, shipping effectiveness, and revenue trends.

This data-driven approach helps organizations make informed decisions in pricing, inventory management, customer engagement, and operational efficiency.

PROBLEM STATEMENT

E-commerce businesses generate massive volumes of raw sales data daily. However, unstructured datasets often contain inconsistencies, missing values, and redundancy, making analysis unreliable.

Without proper cleaning and modelling:

- Sales trends remain unclear
- Product performance is hidden
- Customer behaviour is difficult to track
- Financial metrics become inconsistent

This project resolves these issues by cleaning, validating, and normalizing sales data into structured SQL tables to enable accurate analytics.

E-COMMERCE SALES PERFOMANCE ANALYSIS

I. CURRENT SITUATION

- Raw ecommerce data contains duplicates and inconsistent formats
- Financial calculations are not validated
- Customer and product data are repeated across records
- Business insights are difficult to extract

II. SOLUTION

- Clean data using Power Query
- Normalize dataset into relational tables
- Apply SQL joins and aggregations
- Validate financial calculations
- Enable analytics-driven reporting

Key business questions addressed:

- Calculate the total revenue generated each month**.**
- Which categories contribute most to total profit?
- Which products are receiving poor customer ratings**?**
- Which cities drive the most sales revenue?

DATA DESCRIPTION

Title: E-Commerce Sales Analytics Dataset

- **Data Size:** ~6000+ rows
- **Data Types:** Numeric, Text, Date

Key Fields

- Order ID
- Customer ID
- Product ID
- Quantity
- Discount
- Revenue
- Profit
- Shipping details
- Review ratings

DATA CLEANING STEPS

• Identified and handled missing or null values in key transactional fields such as **quantity**, **unit price**, **discount**, **profit**, and **customer details** to maintain data accuracy.

• Standardized inconsistent text formats by correcting capitalization and spelling variations in **product names**, **categories**, **payment methods**, and **shipping modes**.

• Corrected improper data types by converting fields into appropriate formats — numeric values for financial calculations and date formats for order and shipping timelines.

• Removed duplicate transaction records to ensure each order and customer entry remained unique and reliable.

• Trimmed extra spaces and cleaned hidden characters to improve text consistency across all categorical fields.

• Validated financial calculations by confirming that total sales followed expected formulas and correcting mismatched values.

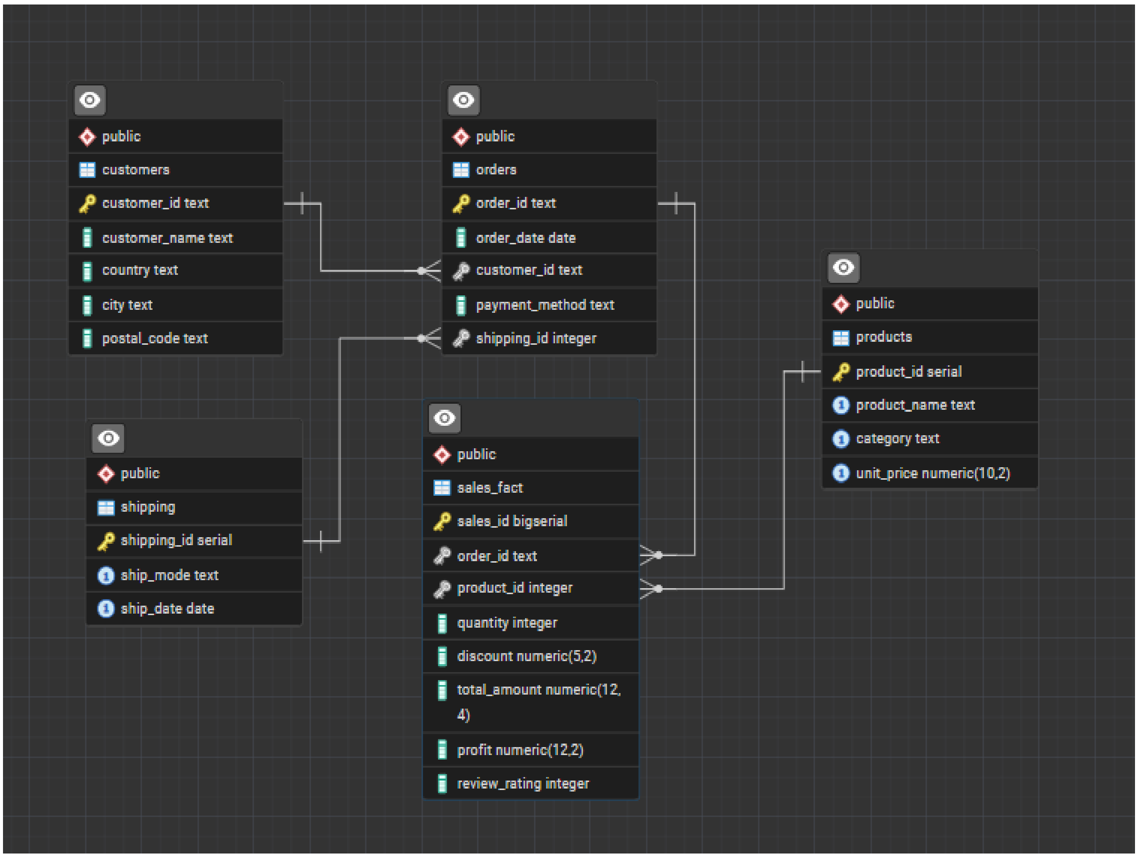
• Filtered invalid or unrealistic entries such as negative quantities or incorrect pricing to preserve logical integrity.

• Prepared the cleaned dataset for normalization by structuring information into logical entities such as customers, products, orders, shipping, and sales transactions.

• Organized the dataset into relational tables to eliminate redundancy and improve SQL query performance

• Normalized the dataset into multiple related SQL tables:

o **customers** *(stores unique customer master information)*
o **products** *(maintains unique product and category data)*
o **orders** *(contains order-level details and customer references)*
o **shipping** *(stores shipment mode and delivery information)*
o **sales_fact** *(captures detailed transaction-level sales data)*



The data model follows a **one-to-many relationship structure**, where each master table (**customers**, **products**, and **shipping**) is linked to multiple transactional records through the **orders** and **sales_fact** tables. Each customer can place multiple orders, and each order can contain multiple product transactions recorded in the sales_fact table.

The foreign keys in the **orders** and **sales_fact** tables reference the primary keys of their respective master tables, ensuring proper linkage between entities. This relational structure simplifies complex analytical queries, improves query performance, and maintains strong data integrity by preventing orphan or inconsistent transaction records.

Category 1 — Revenue & Sales Performance

1. *How much overall revenue and profit has the business generated?*

```
SELECT
  ROUND(SUM(f.total_amount), 2) AS total_revenue,
  ROUND(SUM(f.profit), 2)       AS total_profit
FROM sales_fact f;
```

	total_revenue numeric	total_profit numeric
1	29262737.11	4028156.35

Insight

- Shows overall business performance and total earnings.
- Indicates how effectively sales are converted into profit.
- Helps evaluate yearly or quarterly financial success.
- Supports budgeting, forecasting, and strategic planning decisions.

2. *Calculate the total revenue generated each month*

```
SELECT
  DATE_TRUNC('month', o.order_date) AS month,
  ROUND(SUM(f.total_amount), 2)     AS revenue
FROM sales_fact f
JOIN orders o ON o.order_id = f.order_id
GROUP BY 1
ORDER BY 1;
```

	month timestamp with time zone	revenue numeric
1	2023-01-01 00:00:00+05:30	1998661.20
2	2023-02-01 00:00:00+05:30	2785805.69
3	2023-03-01 00:00:00+05:30	1310843.66
4	2023-04-01 00:00:00+05:30	3141926.70
5	2023-05-01 00:00:00+05:30	2495887.87
6	2023-06-01 00:00:00+05:30	2853556.80
7	2023-07-01 00:00:00+05:30	2432911.98
8	2023-08-01 00:00:00+05:30	1762223.71
9	2023-09-01 00:00:00+05:30	2140086.35
10	2023-10-01 00:00:00+05:30	2540844.66
11	2023-11-01 00:00:00+05:30	2787865.33
12	2023-12-01 00:00:00+05:30	2989160.15
13	2024-01-01 00:00:00+05:30	22963.02

Insight

- Reveals monthly sales performance trends.
- Helps identify peak and low revenue periods.
- Supports seasonal planning and inventory decisions.
- Assists management in forecasting future sales.

3. *What are the top 10 profit-generating products?*

```sql
SELECT
  p.product_name,
  p.category,
  ROUND(SUM(f.profit), 2) AS total_profit
FROM sales_fact f
JOIN products p ON p.product_id = f.product_id
GROUP BY 1,2
ORDER BY total_profit DESC
LIMIT 10;
```

	product_name text	category text	total_profit numeric
1	Air Fryer	Home & Kitch...	228567.83
2	Biography	Books	225929.62
3	Poetry Collecti...	Books	217412.01
4	Laptop	Electronics	216916.70
5	Science Fiction	Books	178510.58
6	Jacket	Clothing	172106.08
7	Hair Serum	Beauty	169947.87
8	Sunscreen	Beauty	145358.39
9	Power Bank	Electronics	142576.92
10	Running Shoes	Clothing	133049.21

Insight

- Identifies products that generate the highest profit.
- Helps prioritize high-margin items for marketing and stocking.
- Supports pricing and product strategy decisions.
- Highlights key contributors to overall business profitability.

4. *Which categories contribute most to total profit?*

```sql
SELECT
  p.category,
  ROUND(SUM(f.profit), 2) AS profit
FROM sales_fact f
JOIN products p ON p.product_id = f.product_id
GROUP BY 1
ORDER BY profit DESC
LIMIT 5;
```

category text	profit numeric
Books	1039500.37
Clothing	703173.23
Beauty	682198.64
Electronics	674274.34
Home & Kitch...	554502.32

Insight

- Highlights product categories that generate the highest profit.
- Helps focus marketing and inventory efforts on profitable segments.
- Supports strategic decisions on category expansion or optimization.
- Reveals where the business earns the strongest returns.

5. *What are the top 10 revenue-generating products?*

```
SELECT
  p.product_name,
  p.category,
  ROUND(SUM(f.total_amount), 2) AS revenue
FROM sales_fact f
JOIN products p ON p.product_id = f.product_id
GROUP BY 1,2
ORDER BY revenue DESC
LIMIT 10;
```

	product_name text	category text	revenue numeric
1	Science Fiction	Books	1735569.98
2	Poetry Collecti...	Books	1484365.19
3	Laptop	Electronics	1436171.39
4	Sunscreen	Beauty	1388409.15
5	Power Bank	Electronics	1104490.71
6	Air Fryer	Home & Kitch...	1085192.47
7	Biography	Books	959441.77
8	Hair Serum	Beauty	933331.89
9	Face Wash	Beauty	898602.74
10	Microwave Oven	Home & Kitch...	876331.58

Insight

- Identifies products that generate the highest sales revenue.
- Helps prioritize top-selling items for marketing and inventory planning.
- Supports revenue-focused product strategy decisions.
- Highlights key drivers of overall sales performance.

Category 2 — Customer Behavior & Value Analysis

6. *Who are the repeat customers based on multiple orders?*

```
SELECT
  c.customer_id,
  c.customer_name,
  COUNT(DISTINCT o.order_id) AS orders_count
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
GROUP BY 1,2
HAVING COUNT(DISTINCT o.order_id) > 1
ORDER BY orders_count DESC;
```

	customer_id [PK] text	customer_name text	orders_count bigint
1	CUST-426	Ivan Ivanov	29
2	CUST-199	Chloe Martin	28
3	CUST-266	Luc Lefebvre	28
4	CUST-210	Kwame Boateng	28
5	CUST-148	Bjorn Borg	28
6	CUST-393	Olga Smirnova	27
7	CUST-164	Hiroshi Tanaka	26
8	CUST-203	unknown	26
9	CUST-355	Ishaan Gupta	26
10	CUST-429	Amara Okafor	25
11	CUST-326	Bjorn Borg	25
12	CUST-208	Wang Wu	25
13	CUST-101	Wolfgang Schmidt	24
14	CUST-316	Omar Farooq	24
15	CUST-374	Matías Villalba	24
16	CUST-292	Yuki Ito	24
17	CUST-395	Joao Silva	23
18	CUST-166	Giovanni Ricci	23
19	CUST-415	Bjorn Borg	23
20	CUST-325	Zahra Diop	23
21	CUST-327	Vihaan Patel	23

Insight

- Identifies customers who make repeated purchases.
- Indicates strong customer loyalty and engagement.
- Helps design retention and reward programs.
- Supports strategies to increase repeat business.

7. Who are the highest value customers based on total spending?

```sql
SELECT
  c.customer_id,
  c.customer_name,
  ROUND(SUM(f.total_amount), 2) AS lifetime_value
FROM sales_fact f
JOIN orders o   ON o.order_id = f.order_id
JOIN customers c ON c.customer_id = o.customer_id
GROUP BY 1,2
ORDER BY lifetime_value DESC
LIMIT 10;
```

	customer_id [PK] text	customer_name text	lifetime_value numeric
1	CUST-107	Alessandro Rossi	707014.00
2	CUST-306	Aisha Mansour	666569.41
3	CUST-393	Olga Smirnova	559864.82
4	CUST-133	Hiroshi Tanaka	520081.33
5	CUST-284	Omar Farooq	504748.49
6	CUST-406	Zahra Diop	478295.33
7	CUST-455	Petra Wagner	476275.22
8	CUST-125	Ricardo Oliveira	456274.04
9	CUST-325	Zahra Diop	454911.25
10	CUST-397	Giovanni Ricci	442320.39

Insight

- Identifies customers who contribute the highest overall spending.
- Highlights high-value customers important to business revenue.
- Supports targeted marketing and loyalty strategies.
- Helps prioritize customer relationship management efforts.

Category 3 — Product Quality & Customer Feedback

8. *Which products are receiving poor customer ratings?*

```
SELECT
  p.product_name,
  p.category,
  ROUND(AVG(f.review_rating)::numeric, 2) AS avg_rating,
  COUNT(*) AS ratings_count
FROM sales_fact f
JOIN products p ON p.product_id = f.product_id
WHERE f.review_rating IS NOT NULL
GROUP BY 1,2
HAVING AVG(f.review_rating) < 3
ORDER BY avg_rating;
```

	product_name text	category text	avg_rating numeric	ratings_count bigint
1	Coffee Maker	Home & Kitch...	2.71	129
2	Tennis Racket	Sports	2.93	138
3	Toaster	Home & Kitch...	2.96	121
4	Power Bank	Electronics	2.99	144
5	Wireless Headphon...	Electronics	2.99	127

Category 4 — Geographic Sales Analysis

9. *Which cities drive the most sales revenue?*

```
SELECT
  c.country,
  c.city,
  ROUND(SUM(f.total_amount), 2) AS revenue
FROM sales_fact f
JOIN orders o    ON o.order_id = f.order_id
JOIN customers c ON c.customer_id = o.customer_id
GROUP BY 1,2
ORDER BY revenue DESC
LIMIT 10;
```

	country text	city text	revenue numeric
1	Germany	Hamburg	1766474.62
2	Germany	Berlin	1588339.64
3	India	Chennai	1106187.74
4	France	Nice	985471.74
5	United Stat...	Miami	960986.46
6	United Stat...	San Francis...	947987.42
7	United Stat...	Dallas	946629.53
8	France	Toulouse	940764.72
9	Canada	Ottawa	936230.31
10	United Stat...	unknown	913953.10

Insight

- Identifies cities that generate the highest sales revenue.
- Reveals strong geographic markets for the business.
- Supports location-based marketing and expansion decisions.
- Helps optimize inventory and logistics planning by region.

Category 5 — Discount & Pricing Impact

10. *How do different discount ranges impact total revenue and profit?*

```
SELECT
  CASE
    WHEN f.discount = 0 THEN '0%'
    WHEN f.discount <= 0.10 THEN '1-10%'
    WHEN f.discount <= 0.20 THEN '11-20%'
    ELSE '21%+'
  END AS discount_bucket,
  ROUND(SUM(f.total_amount), 2) AS revenue,
  ROUND(SUM(f.profit), 2)       AS profit
FROM sales_fact f
GROUP BY 1
ORDER BY 1;
```

	discount_bucket text	revenue numeric	profit numeric
1	0%	8419689.66	1093284.64
2	1-10%	7418685.13	1085334.56
3	11-20%	6358825.21	815476.73
4	21%+	7065537.11	1034060.42

Insight

- Shows how different discount levels influence revenue and profit.
- Helps evaluate the effectiveness of pricing and promotional strategies.
- Identifies discount ranges that balance sales growth and profitability.
- Supports smarter decision-making on future discount policies

Key insights and Recommendations

Category 1 — Revenue & Sales Performance

Insights

- Revenue is driven by a small group of high-performing products and categories.
- Monthly sales trends reveal clear peak and low periods.
- Profit contribution varies significantly across products.
- Certain categories consistently generate strong returns.

Recommendations

- Focus marketing efforts on top-performing products.
- Ensure high-demand items remain well stocked.
- Adjust pricing strategies for low-profit products.
- Use peak sales periods for targeted promotional campaigns.

Category 2 — Customer Behavior & Value Analysis

Insights

- Repeat customers indicate strong loyalty and engagement.
- A small group of customers contributes a large portion of total spending.
- Customer purchasing behavior reveals opportunities for retention strategies.

Recommendations

- Introduce loyalty programs for repeat customers.
- Offer personalized promotions to high-value customers.
- Strengthen customer engagement strategies to increase repeat purchases.

Category 3 — Product Quality & Customer Feedback

Insights

- Some products receive consistently low ratings.
- Customer feedback highlights potential quality or expectation gaps.
- Poor ratings may impact long-term product performance.

Recommendations

- Investigate low-rated products for quality improvements.
- Improve product descriptions and presentation.
- Monitor customer feedback regularly to maintain satisfaction.

Category 4 — Geographic Sales Analysis

Insights

- Certain cities generate significantly higher revenue.
- Geographic sales patterns indicate strong market concentration.

Recommendations

- Focus regional marketing on high-performing locations.
- Optimize logistics and inventory distribution by region.
- Explore expansion opportunities in similar markets.

Category 5 — Discount & Pricing Impact

Insights

• Discounts influence customer purchasing behavior.
• Higher discounts increase sales volume but may reduce profit margins.
• Moderate discount levels balance revenue and profitability.

Recommendations

• Apply discounts strategically rather than universally.
• Focus promotions on price-sensitive products.
• Monitor profit impact before launching discount campaigns.

Conclusion

This e-commerce sales analytics project demonstrates how cleaned and normalized transactional data can be transformed into meaningful business intelligence using SQL. By analyzing revenue trends, product profitability, customer behavior, geographic performance, and discount impact, the project provides a clear view of the company's operational and financial performance.

The structured database model ensures data accuracy, reduces redundancy, and enables efficient querying, making large datasets easier to analyze. The insights generated support informed decision-making in areas such as pricing strategy, inventory planning, customer retention, and market expansion.

Overall, this project highlights the practical value of data cleaning, normalization, and SQL analytics in driving smarter business strategies and improving long-term profitability.